Executive Summary

The present study is about analyzing the factors that can help us in predicting whether a topic in a discussion forum will be closed or not in the future. To analyze this we use our data set from a famous discussion forum ScienceForums.Net (SFN). We observe the variables on per day basis and use them to analyze with what probability a topic is going to be closed.

Based on exploratory data analysis and comparing different logistic regression models using the likelihood ratio test we came up with the model shown in equation (a)

$$\begin{aligned} \log \frac{P(state=open)}{1-P(state=open)} \sim & author.banned + posts + views + authors \\ & + deleted.posts + posts \cdot authors + category \end{aligned} \tag{a}$$

We have verified our model using cross validation and the Hosmer-Lemeshow test for Goodness-of-Fit. The model has the accuracy of test accuracy of about 96 % but prone to high false-positive as shown in Table I as classes in the available data is highly imbalanced.

Predicted	**Actual**		**Accuracy in percentage**
	open	**closed**	
open	1442	55	96.33%
closed	0	3	

Table I: Confusion matrix for verifying the model on test set. Shows High accuracy but percentage of false postive is very high due to imbalnce of class in training set

Finally, we came up with the conclusion that the number of posts per day, the number of authors per day, the field of topic discussion and whether or not the topic starter is banned determine whether a topic is going to be closed or not. For example, ff topics are from Religion and Philosophy, Speculations or General Science then there is a high chance that the topic will be closed.

Based on coefficients we conclude that, if there is a large number of posts per day for the same topic it indicates that the topic may be closed in the future. A similar pattern has been observed for the number of unique authors per day that are discussing this topic. If a large number of authors per day are participating in discussion then it hints that the discussion may be getting aggressive and the topic may be closed

Analysing factors leading to deletion of topics in discussion forums

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1 Introduction

A famous discussion forum ScienceForums.Net (SFN), which has been open since 2002 has about 800,000 posts. The forum hosts a range of discussion topics from various fields, including standard science topics like physics and chemistry but also topics related to politics, religion and general discussion. Its members include field experts as well as the general audience. Sometimes the discussions get abusive or become personal. To maintain the decorum at the forum there is a team of volunteer moderators who keep discussions from getting out of hand, deleting posts or closing discussions as necessary. The moderator needs to know which topic can go out of hand without manually going through every post of each topic. The ability to correctly predict which topic needs attention most and close before it goes out of hand can increase the efficiency of the moderator in maintaining the decorum of the forum. There is a need to find which factors can efficiently help in determining whether the topic should be under surveillance or not, and close if required. In the current study, we discuss factors that can lead to a topic getting closed. For this, we will test the following hypothesis

- Whether the number of posts per day affect the probability of topic getting closed
- Whether the number of unique authors participating per day affect the probability of topic getting closed
- Whether the number of deleted posts per day affect the probability of topic getting closed

Since we want to predict the chance of topic being closed as it's discussion is going on, it is more appropriate to consider the rate of these variables rather than variables itself. We don't know at what point any topic gets out of control, hence rate will give a better idea. For example, it is possible that if the number of posts per duration is very high, it may indicate that some heated argument is taking place. Therefore, instead of considering absolute values of the number of posts, the number of unique authors, views, and number of deleted posts we will consider the number of posts per day, number of unique authors per day, views per day and number of deleted posts per day.

2 Data Summary

We obtained data set from (Fenner 2012). It contains 10,000 samples of discussion topics. Table 2.1 shows the details provided about each topic in the given data set. Here, the outcome variables is **state**

Table 2.1: Details of each topic provided. Each variable provides certain details about the topic as shown in description.

Variable	**Description**
tid	Topic ID
state	"closed" indicates the topic is visible but no longer open for discussion.
posts	Number of posts which have not been deleted.
views	Number of times the topic has been viewed.
duration	Number of days elapsed between the first and last posts.
startdate	Date and time the topic was started.
category	Topic belongs to one of the following forum : 1. Homework help 2. Physics 3. Chemistry 4. Philosophy and religion 5. Medicine 6. Engineering 7. Miscellaneous sciences 8. Mathematics 9. Speculations 10. General discussion 11. Biology 12. Politics 13. Earth science
authors	Number of distinct authors participated in the topic.
deletedṗosts	Number of posts in the topic deleted.
not_deleted	Whether or not the entire topic has been deleted from view.
pinned	A pinned will always appear at the top of the list of discussions.
author_exp	The number of days the topic starter was registered before starting the discussion.
author.banned	Records if the topic starter is currently banned.

**** Note: Original values of variables, posts, views, authors and deletedṗosts are transformed as per section 2.2**

2.1 Handling missing data

In the given data, there are some of the topics which were not part of any of the given category in 2.1, hence we have assigned them a separate category called “Unknown”.

2.2 Transforming values of posts, views, authors and deleted posts

We obtained posts per day, unique authors per day, views per day and deleted.posts per day by dividing variables with corresponding topic's duration. There are some topics with 0 duration. Which means data for these topics was collected on the same day when they were created. In this case, we considered the absolute value of those variables as there rates. Further, histogram plots showed the distribution of these variables to be highly skewed, hence we have considered the logarithmic transformation of these variables. Later in our discussion whenever we are mentioning these variables, we are discussing the logarithmic transformation of their rates.

3 Exploratory Data Analysis

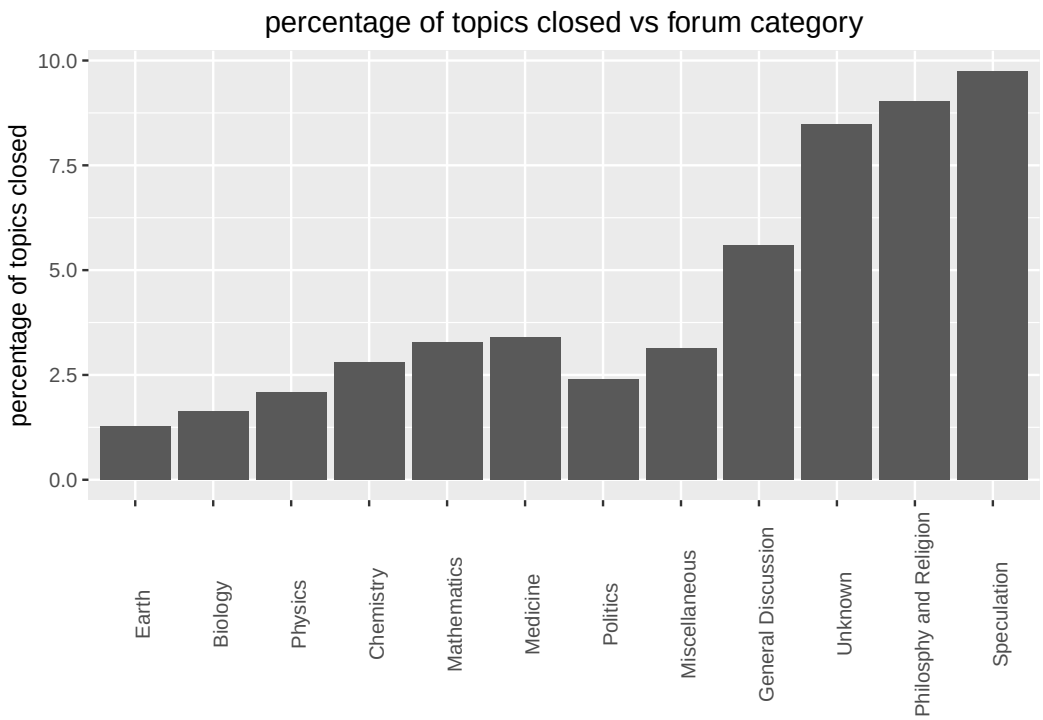

Figure 3.1: Plots to compare category of topic with percentage of topics closed. Topics from categories left of Miscellaneous are less prone to being closed as compared to topics on the right

3.1 Handling outliers

Most of the topics from category Chemistry , Unknown had disproportionately large number of views for low number of posts and number of authors. It did not fit the patterns as shown in 3.2. We have removed these topics from our data.

3.2 Observed relationship among variables

From figure 3.1 we observe, topics from standard academic fields such as Earth, Physics, Chemistry, Mathematics, Politics which involves mostly experts are less prone to deletion then topics from the category such as General Discussion, Unknown and Speculations which involves mostly non experts. This suggests that the topics category plays an important role in predicting whether a topic will be closed. Hence, we considered the relation between the variable 'category' and 'state' in our model.

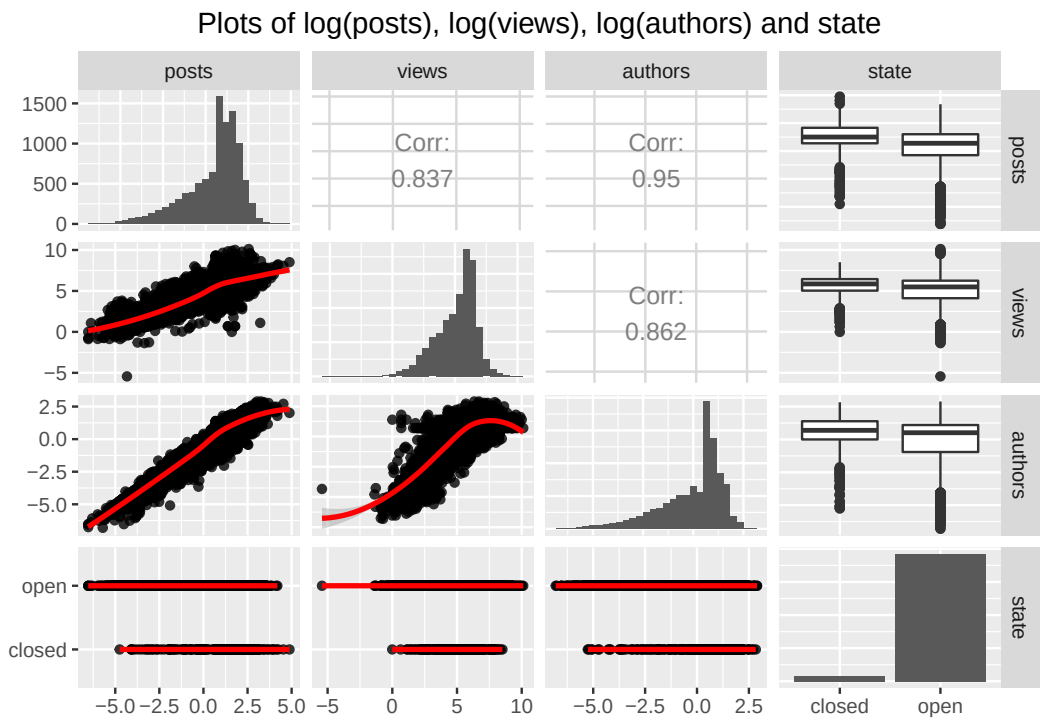

Figure 3.2: Relationship among variables posts, views , authors and state after removing outliers. A strong correlation exist between author, views and posts. More number of posts per day and views per day and authors per day indicates towards topic being closed. Classes in data are highly skewed.

Figure 3.2 shows a strong positive correlation between a number of views, posts and

unique authors per day. The post is closed. suggesting that discussion went out of control among very few authors. Overall, there is a linear relationship between all three of the variables. The rightmost column of figure 3.2 suggests that a large number of views, posts or authors per day leads to topics getting closed.

From the data, we observed that 404 remained open and 54 topics (approx. 13.34%) were closed with there topic starter being were banned. Similarly, 7243 topics remained open while just 186 (approx. 2.56%) were closed when their topic starter was not banned. It suggests the chance of a topic getting closed with the topic starter being banned is high, hence we considered variable author.banned in our model.

For most of the topics, it was observed that a large number of posts getting deleted did not lead to there closing . Large number of deleted posts are still not closed. There must be some more factors playing role in this decision to verify we have considered a relationship of the not_deleted with the 'sate' variable. The coefficient is supposed to be nonsignificant.

Apart from the discussed case, we did not find any relation to other variables with the state. We have suggested three models as discussed in the next section.

4 Method

Based on exploratory data analysis we suggest causal diagram as shown in 4.1.

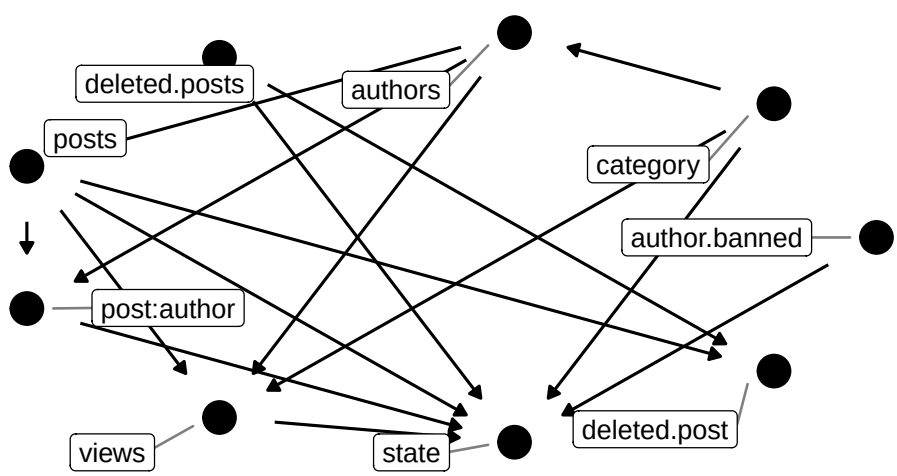

Figure 4.1: Direct Acyclic graph showing relationship among variables with state as outcome variable

For our study, views', 'authors' 'posts' and 'deleted.posts' are exposure variables

and ‘state’ is the outcome variable. Based on causal paths, we come up with two different logistic regression model as shown in equation 4.1 and equation 4.2. We have also used a third model which is a combination of model 1 and model 2 as shown in equation 4.3. We have divided our 10000 samples into 8500 training and 1500 test set. We estimate the coefficient of our models using the training set.

$$\begin{aligned}\log \frac{P(state=open)}{1-P(state=open)} \sim & author.banned + posts + views + authors \\ & + deleted.posts + posts \cdot authors\end{aligned} \tag{4.1}$$

$$\begin{aligned}\log \frac{P(state=open)}{1-P(state=open)} \sim & author.banned + posts + views + deleted.posts \\ & + posts \cdot authors + category\end{aligned} \tag{4.2}$$

$$\begin{aligned}\log \frac{P(state=open)}{1-P(state=open)} \sim & author.banned + posts + views + authors \\ & + deleted.posts + posts \cdot authors + category\end{aligned} \tag{4.3}$$

Table 4.1 shows the estimated values are the odds ratios for each variable. In case of posts, views, authors and deleted_notes if there values scales by b then the odd ratio of state=“open” will scale by $b^{\text{coefficient value}}$. In case of variables, “category” and “author.banned” it represents by odd ratio will scale by $\exp^{\text{coefficient value}}$. The ’Constant’ variable’s coefficient will be the odd ratio if all other variables were close to 0.

In Table 4.1 from model 1 we observe that posts, authors, authors.banned, the interaction of posts and authors is significant. If the topic starter is banned, the likelihood of the topic to remain open decreases. If posts per day and unique author per day increase, the likelihood of topic to remain open also decrease. The interaction between posts and authors suggests that if there are a large number of unique authors and if posts per day increase, it increases the likelihood of topic to remain open. Also, if the number of deleted.posts per day increases the odds ratio decreases, Although the variable is not that significant. The number of views does not seem to have any effect on the odds ratio. A similar conclusion can be drawn from the model 2 for variables posts, author.banned, deleted.posts and interaction term. For any forum category other than Biology and Homework help, the odd ratio decreases. Similar conclusions can be drawn from the Model 3 for all variables.

Based on the loglikelihood ratio test we find the Model 3 to be the best model of all three. Hence, we choose this model and discuss its performance in the later section. The p-values in Table 4.1 are calculated using Wald test. It suggests that in Model 3 posts, authors, authors.banned and forum category are significant whereas views and deleted posts are not significant.

Table 4.1: Comparison of all three models. Based on the Logliklihood ratio test we choose Model 3 as our final model. Vairables like category, posts, authors and author.banned are most significant based on Wald test

	Coefficient of three logistic regression models		
	Model 1	Model 2	Model 3
	(1)	(2)	(3)
author.banned	-1.53^{***}	-1.33^{***}	-1.34^{***}
posts	-0.11^{***}	-0.11^{***}	-0.10^{***}
views	0.0002	0.0001	0.0001
authors	-0.10^{***}		-0.09^{**}
deleted.posts	-0.02	-0.05	0.02
category.Chemistry		-0.61^{*}	-0.59
category.Earth		-0.0000	-0.001
category.Engineering		-0.21	-0.19
category.General Discussion		-1.08^{***}	-1.00^{***}
category.Homework		0.76	0.77
category.Mathematics		-0.82^{**}	-0.83^{**}
category.Medicine		-0.84^{**}	-0.84^{**}
category.Miscellaneous		-0.67^{*}	-0.65^{*}
category.Philosphy and Religion		-1.37^{***}	-1.39^{***}
category.Physics		-0.08	-0.06
category.Politics		-0.10	-0.07
category.Speculation		-1.38^{***}	-1.36^{***}
categoryUnknown		-1.58^{***}	-1.56^{***}
posts:authors	0.01^{***}	0.01^{***}	0.01^{***}
Constant	3.92^{***}	4.51^{***}	4.61^{***}
Observations	8,497	8,497	8,497
Log Likelihood	-1,278.21	-1,228.91	-1,225.93
Akaike Inf. Crit.	2,570.42	2,495.83	2,491.86
Residual Deviance	2,556.42	2,457.83	2,451.86
Null Deviance (df = 8496)	2,752.32	2,752.32	2,752.32
Note:			*p<0.1; **p<0.05; ***p<0.01

Table 4.2: Cross validation for each fold. Accuracy remains same, suggesting model trained correctly without bias

Name	Fold1	Fold2	Fold3	Fold4	Fold5	Fold6	Fold7	Fold8	Fold9	Fold10
Acc.	96.1	96.4	96.9	96.1	97.1	96.1	95.2	95.5	95.8	96.8

4.1 Model Diagnostic

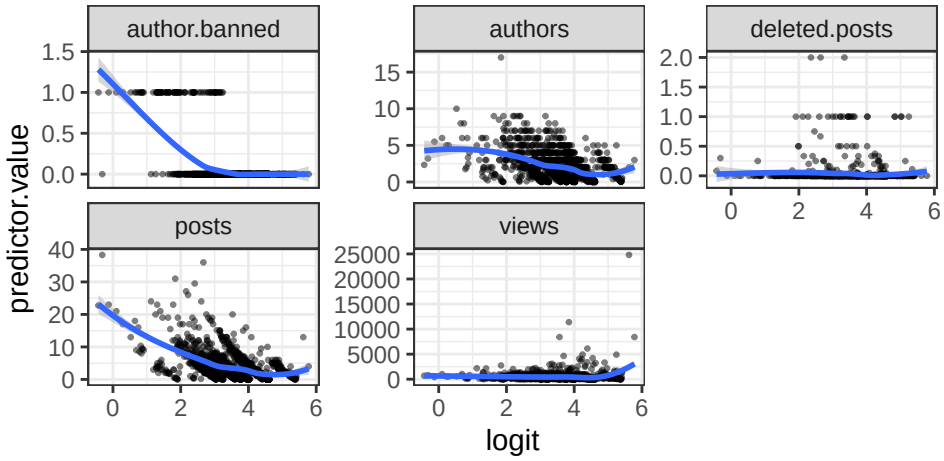

Figure 4.2: Smoothed scatter plots show that variables author.banned, authors, views, posts and deleted.posts are all quite linearly associated with the state outcome in logit scale

Predicted	**Actual**		**Accuracy in percentage**
	open	**closed**	
open	1442	55	96.33%
closed	0	3	

Table 4.3: Confusion matrix for verifying Model 3 on test set. Shows High accuracy but percentage of false postive is very high due to imbalnce of class in training set

Table 4.2 shows that cross validation error is consistent with a standard deviation of error as 0.61 which is low, suggesting the model accuracy does not vary with the training set. It has on an average error of 3.8% which is low suggesting low bias. This error matches almost with test error. The model's predicted values almost fit with all the variables on the logit scale, suggesting that the variables considered were correct. In table 4.3 although the error is low, since the dataset has an unbalanced class, it is suffering from high false-positive error. Using Hosmer-Lemeshow test for Goodness-of-Fit we get p-value 0.73, which suggests a good overall fit for our model.

5 Conclusion

A logistic regression model was suggested to study the effect of the number of posts, views and unique authors per day. We obtained around 96% accuracy. Based on the

Wald significance test we found that posts and unique authors significantly affect the chances of post getting closed however their no such effect of the number of views. Based on the author.banned's coefficient value it has significant impact on the probability of the topic getting closed. Variable category plays a significant role in such a prediction. Mostly if the field of the topic is nonscience based it can get closed with high probability

During our analysis, we observed views is a strong correlation with posts and does not vary with the forum. Some of the topics such as chemistry have a disproportionately large number of views, although it was expected from forums like Speculation or Politics.

We also observed that if a few people are involved in a large number of posts it suggests that this post will be deleted. This is also satisfied with the coefficient values of posts, authors and post:authors obtained from our final model.

Based on the Wald test number of deleted posts does not affect the topic getting closed. that pinned topic does not seem to have any effect on the number of views.

The author_exp does not seem to have any effect on the variable views, posts or state. Most of the discussions are started by newcomers which were not closed.

We observed some weak relationships of posts and views with the number of the deleted post. Finally, we did not observe any relationship between topics being deleted and topics being closed.

The class distribution in the data set is highly unbalanced. This has lead to a very high false-positive error. Having more datasets with state="open" will certainly help. If more data cannot be arranged then we can consider a logistic regression model where we assign weights to each sample as per there class. This can improve performance. But overall, we believe even after improving performance, the relationship between and state and other variables will not change significantly although it may change coefficient values.

6 References

Fenner, Martin. 2012. "One-Click Science Marketing." *Nature Materials* 11 (4): 261–63. https://doi.org/10.1038/nmat3283.